

Rolling Admissions MDP

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June 2026

1 Setup and Assumptions

The system operates in discrete time on a finite horizon $t \in \{1, 2, \dots, T\}$. We discretize the student population using two parameters:

1. **Quality Tier:** $i \in \{1, \dots, r\}$, where $q_i = i/r$ represents the normalized quality of a student in tier i .
2. **Attribute Combination:** $a \in \{1, \dots, A\}$, representing unique cross-products of diversity attributes (e.g., gender, nationality, socioeconomic status).

Students are partitioned into $r \times A$ distinct buckets. To preserve high-fidelity behavioral tracking without losing computational efficiency, we track the system state using three primary multi-dimensional matrices:

- **Waitlist** ($\mathbf{N}_t \in \mathbb{N}_0^{r \times A \times \tau_{\max}}$): Counts of applicants who have applied but have not yet been extended an offer, partitioned by tier i , attribute combination a , and the number of periods the applicant has spent on the waitlist $\tau \in \{1, \dots, \tau_{\max}\}$.
- **Matriculated Students** ($\mathbf{M}_t \in \mathbb{N}_0^{r \times A}$): Counts of students who have received an offer and accepted it, tracked by tier i and attribute combination a .
- **Outstanding Offers** ($\mathbf{O}_t \in \mathbb{N}_0^{r \times A \times W}$): Counts of pending offers, partitioned by tier i , attribute combination a , and remaining decision periods $w \in \{1, \dots, W\}$.

We assume the initial state $\mathbf{N}_0 = \mathbf{M}_0 = \mathbf{O}_0 = \mathbf{0}$. At any time t , the state of the system is uniquely described by the tuple $S_t = (\mathbf{M}_t, \mathbf{N}_t, \mathbf{O}_t)$. The remaining physical capacity of the incoming class is denoted by $c_t = C - \sum_{i=1}^r \sum_{a=1}^A m_{i,a,t}$, where C is the institution's total capacity target.

2 Time-Dependent Transition Probabilities

To align the model with historical empirical data, all transition and arrival dynamics are generalized to be time-dependent.

- **Applicant Arrivals:** The total number of new arrivals at time t , denoted by E_t , follows a Poisson distribution with mean $\lambda(t)$:

$$E_t \sim \text{Poisson}(\lambda(t))$$

Each arriving applicant is independently assigned to a quality tier i and attribute combination a with probability $\pi_{i,a}(t)$, where $\sum_{i,a} \pi_{i,a}(t) = 1$. The number of arrivals into each specific bucket (i, a) at time t , denoted by $\Delta n_{i,a,t}$, is jointly modeled using a multinomial distribution:

$$(\Delta n_{1,1,t}, \dots, \Delta n_{r,A,t}) \sim \text{Multinomial}(E_t, [\pi_{1,1}(t), \dots, \pi_{r,A}(t)])$$

New applicants always enter the waitlist with an initial tenure of $\tau = 1$.

- **Waitlist Dropouts:** A student of tier i and attribute combination a remaining on the waitlist with tenure τ at time t independently drops out of the admissions process with a tenure-fatigued probability $\delta_{i,a}(t, \tau)$. The number of students who survive attrition and remain available for period $t + 1$ is modeled binomially:

$$n_{i,a,\tau+1,t+1} \sim \text{Binomial}(\tilde{n}_{i,a,\tau,t}, 1 - \delta_{i,a}(t, \tau))$$

- **Outstanding Offer Responses:** Let $w \in \{1, 2, \dots, W\}$ represent the number of periods remaining before an offer's decision deadline. At time t , a student in quality tier i and attribute combination a holding an offer with w periods remaining will execute one of three choices:
 - **Accept** the offer with probability $\theta_{i,a}(t, w)$,
 - **Reject** the offer (or let it expire) with probability $\mu_{i,a}(t, w)$,
 - **Remain undecided** with probability $1 - \theta_{i,a}(t, w) - \mu_{i,a}(t, w)$.

Because students cannot remain undecided once the decision window expires, we enforce the hard boundary constraint:

$$\theta_{i,a}(t, 1) + \mu_{i,a}(t, 1) = 1 \quad \forall i, \forall a, \forall t$$

For all periods with remaining validity ($w > 1$), the probability of remaining undecided is strictly non-negative, implying $\theta_{i,a}(t, w) + \mu_{i,a}(t, w) \leq 1$.

3 Timeline and Action Constraints

Within each period t , events occur in the following precise chronological order:

1. **Arrival Phase:** The system experiences a stochastic influx of new applicants $\Delta n_{i,a,t}$ determined by the Poisson-Multinomial process. We define the intermediate, post-arrival waitlist tensor as $\mathbf{N}'_t$, where arrivals append directly to the first tenure slice:

$$n'_{i,a,\tau,t} = \begin{cases} n_{i,a,1,t} + \Delta n_{i,a,t} & \text{if } \tau = 1, \\ n_{i,a,\tau,t} & \text{if } \tau > 1. \end{cases}$$

2. **Decision Phase:** The school observes the post-arrival state $S'_t = (\mathbf{M}_t, \mathbf{N}'_t, \mathbf{O}_t)$ and selects an admission action matrix $\mathbf{K}_t \in \mathbb{N}_0^{r \times A \times \tau_{\max}}$, where $k_{i,a,\tau,t}$ is the number of offers extended to waitlist applicants in bucket (i, a, τ) . The chosen action must belong to the feasible action space $\mathcal{K}(S_t)$, defined by structural and capacity boundaries:

$$\mathcal{K}(S_t) = \left\{ \mathbf{K}_t \in \mathbb{N}_0^{r \times A \times \tau_{\max}} \mid k_{i,a,\tau,t} \leq n'_{i,a,\tau,t} \ \forall i, a, \tau, \text{ and } \sum_{i,a,\tau} k_{i,a,\tau,t} \leq c_t \right\}$$

The state variables are immediately modified to their intermediate, post-decision values:

- Post-decision waitlist: $\tilde{n}_{i,a,\tau,t} = n'_{i,a,\tau,t} - k_{i,a,\tau,t}$
- Post-decision offers at maximum window: All newly extended offers enter at the full horizon W :

$$\tilde{o}_{i,a,W,t} = o_{i,a,W,t} + \sum_{\tau=1}^{\tau_{\max}} k_{i,a,\tau,t}$$

For all shorter horizons $w < W$, the outstanding offer counts remain unaltered: $\tilde{o}_{i,a,w,t} = o_{i,a,w,t}$.

3. **Stochastic Response Phase:**

- Remaining waitlist students ($\tilde{\mathbf{N}}_t$) independently choose to drop out or remain based on $\delta_{i,a}(t, \tau)$.
- Holders of post-decision outstanding offers ($\tilde{\mathbf{O}}_t$) execute their choices to accept, reject, or remain undecided.

4. **Clock Ticks:** Undecided offers age down by one period of validity, surviving waitlist candidates age by one period of tenure, and the system clock increments from $t \rightarrow t + 1$.

4 Stochastic Transition Dynamics

To simulate the state's transition from time t to $t + 1$, we sample from Binomial and Multinomial distributions based on the post-decision parameters.

4.1 Waitlist Updates

Students remaining on the post-decision waitlist experience attrition. The candidates who do not drop out advance to the next tenure increment $\tau + 1$ for the subsequent period:

$$n_{i,a,\tau+1,t+1} \sim \text{Binomial}(\tilde{n}_{i,a,\tau,t}, 1 - \delta_{i,a}(t, \tau)) \quad \forall \tau \in \{1, \dots, \tau_{\max} - 1\}$$

Note that any student reaching $\tau > \tau_{\max}$ is automatically cleared from the system due to expiration. New arrivals at $t+1$ will populate the $\tau = 1$ boundary slice.

4.2 Outstanding Offer Decisions

For each tier i , attribute combination a , and remaining validity window $w \in \{1, \dots, W\}$, the pool of post-decision outstanding offers $\tilde{o}_{i,a,w,t}$ splits into three disjoint components:

- $A_{i,a,t}^w$: The number of students who accept the offer at time t ,
- $R_{i,a,t}^w$: The number of students who reject (or let expire) the offer at time t ,
- $U_{i,a,t}^w$: The number of students who remain undecided at time t .

These outcomes are jointly simulated via a cell-wise Multinomial draw:

$$(A_{i,a,t}^w, R_{i,a,t}^w, U_{i,a,t}^w) \sim \text{Multinomial}(\tilde{o}_{i,a,w,t}, [\theta_{i,a}(t, w), \mu_{i,a}(t, w), 1 - \theta_{i,a}(t, w) - \mu_{i,a}(t, w)])$$

4.3 State Progression to $t + 1$

Once the stochastic variables are sampled, the components for period $t + 1$ are compiled. The matriculation matrix accounts for all new acceptances across outstanding valid horizons:

$$m_{i,a,t+1} = m_{i,a,t} + \sum_{w=1}^W A_{i,a,t}^w \quad \forall i, \forall a$$

The outstanding offer matrix updates by decrementing the undecided segments down by one validity horizon. For all $i \in \{1, \dots, r\}$ and $a \in \{1, \dots, A\}$, the elements progress as:

$$o_{i,a,w,t+1} = \begin{cases} U_{i,a,t}^{w+1} & \text{if } w < W, \\ 0 & \text{if } w = W. \end{cases}$$

5 Objective Function

The goal of the Markov Decision Process is to find an admission policy that maximizes total expected Net Benefit Z . The terminal reward captures cumulative class quality balanced against structural regularizations penalizing a lack of group diversity and class overbooking.

Let $M_{a,T} = \sum_{i=1}^r m_{i,a,T}$ denote the aggregate count of matriculated students belonging to attribute combination a at the terminal horizon, and let $M_T = \sum_{i=1}^r \sum_{a=1}^A m_{i,a,T}$ equal the total final class size. The Herfindahl-Hirschman Index (HHI) measures representation concentration across diversity attribute combinations. The objective function is formulated as:

$$Z = \underbrace{\sum_{i=1}^r \sum_{a=1}^A m_{i,a,T} \cdot q_i}_{\text{Class Quality}} - \underbrace{\phi \cdot \sum_{a=1}^A \left(\frac{M_{a,T}}{M_T} \right)^2}_{\text{Diversity Penalty (HHI)}} - \underbrace{\psi \cdot \max(0, M_T - C)^2}_{\text{Overbooking Penalty}}$$

where ϕ balances the marginal value of background representation and ψ scales the quadratic cost of violating physical enrollment capacity limitations.